

# M KARTHIKEYAN

AI Engineer | Machine Learning Specialist | OT Security Analytics

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🌐 **Portfolio:** [karthikeyan1708.github.io](https://karthikeyan1708.github.io) — *Interactive portfolio with projects, engagements & experience*

## Objective

AI Engineer with **1.8+ years** at PricewaterhouseCoopers, specializing in production-grade machine learning pipelines for industrial cybersecurity. Seeking opportunities to apply expertise in anomaly detection, large-scale data engineering, and AI-powered threat analysis to build impactful, scalable solutions.

## Education

**B.Tech, Computer Science & Engineering (Artificial Intelligence)** 2020–2024  
*School of AI & Computing, Amrita Vishwa Vidyapeetham, Coimbatore* CGPA: 8.5 / 10  
*Key Coursework:* Machine Learning, Deep Learning, NLP, Computer Vision, Data Structures & Algorithms, Software Engineering

## Experience

**Associate – AI & OT Cybersecurity** Oct 2024 – Present  
*PricewaterhouseCoopers (PwC), Hyderabad* 1.8+ years

### AI-Driven Security Analytics

- Designed and deployed an **ML-based anomaly detection pipeline** processing 500K+ daily OT sensor events, achieving 85% precision and reducing false-positive rates by 30%.
- Engineered Python workflows to ingest and analyze **multi-terabyte PCAP datasets**, extracting 50+ protocol-level features for downstream classification models.
- Developed an **XGBoost risk-scoring engine** that ranked vulnerabilities by exploitability and asset criticality, cutting manual triage time by 40%.
- Built reusable **NLP pipelines** to parse security policy documents and identify compliance gaps against ISO 27001:2022 controls with 90%+ extraction accuracy.

### Assessment Delivery

- Delivered **6 end-to-end OT security assessments** across aviation, energy, telecom, and manufacturing – encompassing network architecture review, stakeholder workshops, and executive reporting.
- Executed on-site audits across **4 geographically distributed sites** under aggressive 3-week timelines, delivering maturity assessments with prioritized remediation roadmaps.
- Performed **firewall rule analysis and network segmentation reviews** for industrial control environments, uncovering 25+ critical misconfigurations per engagement.
- Conducted physical and logical security reviews of telecom infrastructure across 5 tower locations, scripting configuration compliance checks in Python.

### SOC Operations & Continuous Monitoring

- Providing **24/7 shift-based SOC support** for a global industrial technology provider – real-time threat monitoring, alert triage, and incident escalation using Microsoft Sentinel and Defender for IoT.

## Projects

**Automatic Speech Recognition & Emotion Detection – Tamil Language** Apr–May 2024

- Fine-tuned OpenAI's **Whisper Base** model with transfer learning on a custom Tamil-speech corpus; appended bidirectional LSTM layers for temporal emotion modeling, achieving **85% accuracy** across 6 emotion categories.
- Designed a robust data pipeline handling audio augmentation (noise injection, pitch shifting, time stretching), MFCC/spectrogram extraction, and batch processing of 12+ hours of raw audio daily.
- Deployed inference endpoint via FastAPI with sub-200ms latency; containerized with Docker for reproducible deployment.

## Communicable Disease Outbreak Prediction System

Jan–Mar 2023

- Developed an ensemble of **RNN and SVR models** to forecast dengue and chikungunya outbreaks with **92% accuracy** over a 12-month prediction horizon.
- Engineered 30+ features from epidemiological, meteorological, and demographic datasets totaling 2M+ records; applied PCA and recursive feature elimination for dimensionality reduction.
- Built interactive Plotly dashboards for public-health stakeholders, enabling drill-down analysis by region, time period, and disease type.
- **Published** in Springer – *Inventive Communication and Computational Technologies* (Oct 2023).

## Technical Skills

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- **Languages:** Python (Advanced), SQL, MATLAB, Java
- **ML/DL Frameworks:** TensorFlow, PyTorch, Keras, Scikit-learn, XGBoost, Hugging Face, ONNX
- **Data & Visualization:** NumPy, Pandas, Dask, Matplotlib, Seaborn, Plotly, Power BI
- **MLOps & DevOps:** Docker, Git, CI/CD (GitHub Actions), FastAPI, REST APIs, Linux
- **Cloud:** AWS (S3, SageMaker, Lambda), Azure (Sentinel, Blob Storage)
- **Security Tools:** Wireshark, Claroty, Tenable, PCAP Analysis, OT/ICS Protocol Analysis

## Certifications

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**ISO 27001:2022 Lead Implementer** – Information Security Management Systems (ISMS)

## Research Works

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M Karthikeyan et al., “*Communicable Disease Prediction Using Machine Learning and Deep Learning Algorithms*,” *Inventive Communication and Computational Technologies*, pp. 979–992, Springer, Oct 2023.

DOI: [10.1007/978-981-99-5166-6\\_66](https://doi.org/10.1007/978-981-99-5166-6_66)